

ATPL EASA STUDY GUIDE | 090 - COMMUNICATIONS

Chapter 5: Weather Broadcasts & Met Information

Syllabus Source: CAE Oxford (pp. 119-142) | Pass Target: >= 90% AviationExam

1. AUTOMATIC TERMINAL INFORMATION SERVICE (ATIS)

ATIS is the continuous broadcast of recorded non-control aeronautical and meteorological information at high-density aerodromes. Its primary operational purpose is to relieve ATC frequency congestion:

- Identification Designator: Each ATIS broadcast is identified by a phonetic letter from the ICAO alphabet (ALPHA through ZULU). When updated, the designator increments sequentially.
- Crew Acknowledgment: On initial contact with Approach or Tower, flight crew must state receipt of the broadcast code letter and QNH: 'INFORMATION ECHO RECEIVED, QNH 1013'.
- Voice vs D-ATIS: Voice ATIS is broadcast on dedicated VHF channels or VOR voice subcarriers. D-ATIS (Data link ATIS) transmits digital text directly into the aircraft cockpit FMS/ACARS printer.

2. STANDARD ATIS CONTENT & MESSAGE SEQUENCE

Message Element	Exact Broadcast Format Example
1. Aerodrome & Code	'GATWICK INFORMATION FOXTROT'
2. Observation Time	'TIME 1420 UTC'
3. Expected Approach & Runway	'EXPECT ILS APPROACH RUNWAY 26L'
4. Runway Surface Conditions	'RUNWAY WET WET WET'
5. Transition Level	'TRANSITION LEVEL 60'
6. Surface Wind	'WIND 240 DEGREES 15 KNOTS GUSTING 25 KNOTS'
7. Visibility & RVR	'VISIBILITY 4000 METRES, RVR RUNWAY 26L 1200 METRES'
8. Present Weather & Clouds	'MODERATE RAIN, FEW 800 FEET, BROKEN 1500 FEET'
9. Temperature, Dewpoint & QNH	'TEMPERATURE 12, DEWPOINT 10, QNH 1014'
10. Operational Warnings	'WINDSHEAR REPORTED ON FINAL APPROACH'

3. VOLMET, SIGMET & AIRMET BROADCASTS

- VOLMET: Continuous broadcast of routine aerodrome weather reports (METAR), special reports (SPECI), and 9-hour trend forecasts (TAF) for major airports, transmitted over scheduled VHF and HF frequencies. - SIGMET: Warning issued by a Meteorological Watch Office (MWO) concerning en-route weather phenomena which may affect the SAFETY OF ALL AIRCRAFT (e.g. severe turbulence, severe icing, thunderstorm lines, volcanic ash). - AIRMET: Information for low-level flights (below FL 100 or FL 150) regarding weather phenomena not included in regional forecasts (moderate icing, moderate turbulence, widespread IMC).

4. RADIOTELEPHONY WEATHER TERMINOLOGY & CAVOK

[!] EXAM TRAP (AVIATIONEXAM) - The Official Definition of 'CAVOK' (Ceiling And Visibility OK)

The term 'CAVOK' replaces visibility, present weather, and cloud groups when ALL FOUR conditions are met: 1. Visibility is 10 KILOMETRES OR MORE; 2. No cloud of operational significance (NO CLOUD BELOW 5,000 FT or below the highest Minimum Sector Altitude, whichever is greater); 3. NO CUMULONIMBUS (CB) and NO TOWERING CUMULUS (TCU) at any altitude; 4. No significant weather phenomena (no rain, snow, fog, thunderstorm, etc.).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: Which of the following conditions is mandatory to report 'CAVOK' in an aeronautical weather broadcast?**

[A] Visibility 5 km and sky clear [B] Visibility 10 km or more, no clouds below 5,000 ft or MSA, no CB/TCU at any level, and no significant weather [C] Wind calm and temperature above freezing [D] QNH exactly 1013.25 hPa

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. CAVOK requires $\geq$ 10 km visibility, no clouds below 5,000 ft or MSA, no CB or TCU at any altitude, and no significant weather.

Q2: How does an ATIS broadcast identify that new or updated meteorological information has been published?

[A] A tone sounds on the emergency frequency [B] The phonetic letter designator advances to the next sequential letter (e.g. from Bravo to Charlie) [C] The frequency changes [D] The broadcast announces 'Flash update'

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. ATIS messages are identified sequentially by phonetic letters (Alpha through Zulu) with each newly issued observation.

Q3: What type of meteorological broadcast provides warnings of severe en-route weather hazardous to ALL aircraft?

[A] METAR [B] TAF [C] SIGMET [D] ATIS

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [C]. SIGMET is an en-route safety warning issued for phenomena that affect the safety of all aircraft (severe icing/turbulence, volcanic ash).

Q4: What is the primary purpose of a VOLMET broadcast?

[A] Providing aerodrome control tower instructions [B] Providing current METARs, SPECIs, and TAFs for major international aerodromes to aircraft in flight [C] Monitoring ELT distress beacons [D] Broadcasting satellite navigation corrections

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. VOLMET broadcasts surface weather reports and aerodrome forecasts to airborne aircraft over VHF and HF.

Q5: How should a pilot acknowledge receipt of ATIS information 'Golf' on initial radio contact with Approach?

[A] 'We have the weather' [B] 'Information Golf received, QNH 1018' [C] 'Roger Golf' [D] 'Weather checked'

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The pilot must explicitly state the phonetic designator ('Information Golf received') and read back the altimeter setting (QNH).

Q6: In an RVR report, what does the designation 'R26L/P1500' signify?

[A] Runway 26 Left RVR is exactly 1,500 feet [B] Runway 26 Left RVR is greater than 1,500 metres [C] Runway 26 Left is closed for 1,500 seconds [D] Precision approach required

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The prefix 'P' indicates that the Runway Visual Range exceeds the maximum reporting capability of the transmissometer (greater than 1,500 m).